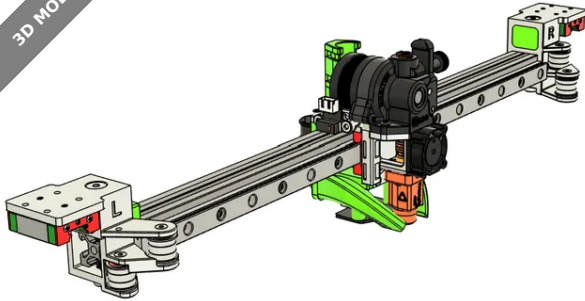


3D MODEL ONLY



Rat Rig V-Core Full Frontal Gantry mod.

M MFBS Design

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Summary

Full Frontal X gantry mod for Rat Rig V-Core 3/3.1

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Tags: [dragon](#) [mod](#) [rat](#) [core](#) [ace](#) [rig](#) [gantry](#) [full](#)
[ratrigvcore](#) [ratrig](#) [ultralight](#) [ratrigvcore3](#) [mfbs](#) [frontal](#)
[ratrigvcore31](#)

The Full Frontal Gantry mod for all models of Rat Rig V-Core 3 and 4 printers, other than IDEX.

Compatibility:

V-Core 3
V-core 3.1
V Core 3 and 3.1 MFBS upgrade
V-Core 3.2h
V-Core 4
V-Core 4 hybrid
OneBox and
Enclosure 2
All Ultralight print heads.

Please note initial testing of this mod has been completed.

Why

1. The Full Frontal Gantry mod dramatically reduces the distance of the nozzle from the Centre of Rotation of the MGN12 carriage in the Y and Z dimensions. reducing nozzle movement due to play/flex in the carriage rail mounting interface to increase printing precision.
2. The The Full frontal mod also dramatically reduces the distance from the belt attachment points from the Centre of Rotation of the MGN12 carriage in both the Y and Z dimensions, reducing the moment of torque applied by the belts to the Centre of Rotation of the MGN12 carriage to increase printing precision.
3. The Ultralight core mounting plate template will provide a simple fixture for a variety of print tools. Print tools could be things like the upcoming Bondtech INDX a pen a drag knife, etc.

Coffee fund

I receive no kick backs or royalties from third party companies such as Funssor and PCBway who manufacture and sell stuff I design and post on Printables.

If you decide to make a donation to my coffee fund (which keeps me up late at night doing more of this stuff) I would appreciate it. There is absolutely no obligation for you to do this. My PayPal account is Liam@FastBikeGear.co.nz

Please remember to post feed back as it is always valuable to me for future iterations.

Many thanks.

Liam Venter

MFBS Design

A STEP file of the Ultralight Core tool head mounting plate is included for those who want to mount other print heads or tool changers.

The core mounting plate is easily modified to support the mounting of a range of tools and print heads. I will release a version of this mounting plate to suit the Bondtech INDX tool changer as soon as Bondtech release the mounting template.

The Ultralight Core tool head mounting plates should be CNC'd or 3D printed in aluminium. PCBway charge about \$26 to print these.

Files for the Dragon Ace Ultralight tool head are included.

PCBway charge about \$32 to print the Ultralight Dragon Ace for Triangle Labs superb Dragon Ace hot end from aluminium.

Files for the Rapido 2 Ultralight tool head are included.

PCBway charge about \$32 to print the Ultralight Dragon Ace for Triangle Labs superb Dragon Ace hot end from aluminium. I highly recommend the use of Funssors optional stand off tube kit with this version of the

Ultralight.

Modular back plate.

I have made the back plate modular.

1. Top (includes umbilical cable mount)
2. CPAP pipe
3. Ducts for part cooling.

Making these three parts modular makes it easier to swap these out for alternative part cooling options.

I have also created drafts of other Ultralight print heads for the following hot ends that I will update when they are finished.

Phaetus Rapido (original)

Phaetus Rapido 2

Triangle Labs Rapido Ace

Peopoly Lancer

Belt Wedge-Lock retaining system.

For the Ultralight tool heads, I have devised a new belt wedge lock retention system.

1. The belts are pushed all the way through curved and tapered slots.,
2. Then a short 8mm length of filament is placed in about the 4th belt groove from the end,
3. The belt is then withdrawn out of the slot until the filament and belt jams in the narrowing taper of the slot.,
4. A small section of belt is left protruding out the slot so that when you want to release the belt you can just pull on the section sticking out with a pair of pliers to release the short piece of filament and then fully withdraw the belt from the slot.,

Pros:

I have tested this and it is extremely secure.

Simple, compact and does not require any screws.

Cons:

A little bit fiddly to fit, but probably quicker than most other systems.

I did experiment with using a toothed wedge instead of a piece of filament, but the filament worked so well the toothed wedge was just an unnecessary complication and easy part to drop on the floor and loose....so is the filament...but you can always find another 8mm long piece of filament! (edited)

Full Metal XY joiners

While all parts of this design have a non commercial license, I may offer a license to one manufacturer (yet to be selected) to offer these for sale as per the XY joiners for my previous V-Core 3 hybrid upgrade.

XY joiners BOM

2 x M5 x 28 Flat Head Cap screws. (Gantry retaining top)

2 x M5 x 10mm Socket Head Cap screws. (Gantry retaining bottom)

7 x M3 x 8mm flat head cap screws (bearing stack retaining)

1 x M3 x 6 mm flat head cap screw for (bearing stack retaining)
4 x Aluminium threaded spindles 14 x 50D x 3mm ID (centre of bearing stack).

Assembly.

XY Joiners:

The two of the bearing stacks mounted in the shorter clevis's must be installed before the gantry is fitted. They can not be installed after the gantry is fitted. These will only fit into place if installed from behind the XY joiners. It is not possible to access the rear of the XY joiners once the gantry is installed.

The other two bearing stacks can be fitted before or after the gantry is installed.

Ultralight:

Depending on the accuracy of the aluminium printed part, which can vary by as much as 0.1mm you may need to fractionally open up the entry point for the belt slots. This can be easily accomplished with a 1.5 mm drill bit.

Updates

8/Sept/2025 increased diameter of heat sink hole in Dragon Ace version of Ultralight print head to make fitting easier.

8/Sept/2025 Increased clearances on XY joiners for belts.

14/Sept/2025 further increased the clearances on the XY joiners for the belts.

15/Sept/2025 Published Rapido 2 Ultralight STL and STEP files.

Still to do:

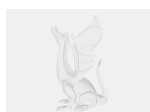
Improved version of the part cooling duct. The improved version will include my normal vertical splitter in each horn of the duct. This should be done in a couple of days.

Ultralight mount for Bondtech INDX.

Model files



rapido-2-ultralight-v24.stl



rapido-2-ultralight-v24.step



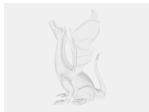
dragon-ace-ultralight-v184.stl



dragon-ace-ultralight-v184.step



ultralight-core-template-v184.stl



ultralight-core-template-v184.step



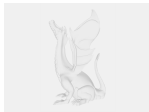
bearing-stack-threaded-tube.stl



bearing-stack-threaded-tube.step



back-plate-top-v99.stl



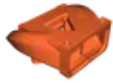
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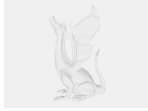
cpap-pipe-v99.stl



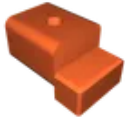
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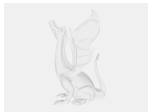
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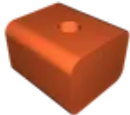
part-cooling-duct-v99.step



x-stop-bumper-v10.stl



x-stop-bumper-v10.step



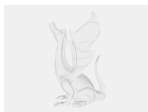
right-xy-joiner-cosmetic-v6.stl



right-xy-joiner-cosmetic-v6.step



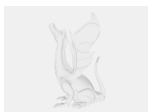
l-front-ffg-xy-joiner-v69.stl



l-front-ffg-xy-joiner-v69.step



r-front-ffg-xy-joiner-v53.stl



r-front-ffg-xy-joiner-v53.step

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